Quantization as Accidental Jailbreak Defense: INT4 Compression Induces Over-Refusal in Susceptible LLMs

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Abstract

Does compressing weights compress safety? We find the opposite—INT4 quantization can induce over-refusal, acting as an accidental jailbreak defense. Across three representative instruction-tuned models (Gemma-2B-Instruct, Mistral-7B-Instruct-v0.2, and Meta-Llama-3-8B-Instruct), we evaluate refusal probabilities using 550 synthetic templates and HarmBench prompts with Llama-3-70B as semantic judge. Capability controls (MMLU/GSM8K) confirm increased refusal is not intelligence loss. While Gemma-2b-it and Meta-Llama-3-8B show no significant change, Mistral-7B shows a trend (uncorrected p=0.027, OR=1.46); after Benjamini-Hochberg FDR correction for 6 comparisons, adjusted p = 0.16, suggesting this is a promising signal warranting larger-scale replication. We release the pipeline, interactive Streamlit dashboard visualizing per-prompt judge labels and refusal margins, and analysis scripts.

1 Introduction

The rapid proliferation of Large Language Models (LLMs) [8] has been accompanied by intensive efforts to align their behavior with human values, typically through supervised fine-tuning and Reinforcement Learning from Human Feedback (RLHF) [1, 6]. Simultaneously, the immense computational cost of deploying these models has popularized weight-quantization techniques, enabling deployment on consumer hardware and significantly improving latency [2, 4].

Despite the widespread adoption of 8-bit and 4-bit precision, the intersection of quantization and safety alignment is critically under-studied. Standard capability benchmarks confirm that perplexity and zero-shot performance degrade gracefully under quantization. However, safety alignment is often brittle; RLHF can act as a shallow "wrapper" over a model's underlying pre-trained distribution. It is an open question whether quantization-induced noise disproportionately disrupts these fine-tuned safety pathways.

In this paper, we systematically evaluate the robustness of LLM refusal behavior under post-training weight quantization. Utilizing Llama-3-70B as a semantic judge over both synthetic limits and HarmBench targets, we find model-family-specific behavioral shifts under INT4 that are independent from capability decay, with Mistral-7B showing a promising trend (adjusted p = 0.16) warranting larger-scale replication.

2 Related Work

Post-Training Quantization. Quantizing LLMs to 8-bit or 4-bit precision reduces memory requirements by mapping continuous weights to discrete, low-precision buckets. Methods like LLM.int8() [2] and GPTQ [4] accomplish this while preserving general zero-shot performance. More

recently, QLoRA introduced 4-bit NormalFloat (NF4) quantization optimized for parameter-efficient tuning [3].

LLM Alignment and Safety. Instruction tuning and constitutional AI paradigms successfully mitigate unsafe outputs [1, 7]. Evaluating this alignment usually involves challenging models with adversarial benchmarks like AdvBench or HarmBench [5]. Prior work has demonstrated that fine-tuning can aggressively overwrite these safety mechanisms (often termed "Shadow Alignment") [10], and that complex "Do Anything Now" (DAN) jailbreaks can reliably bypass them [9]. We extend this line of inquiry by investigating quantization itself as a mechanism that disrupts original alignment bounds.

3 Methodology

3.1 Dataset Generation and External Validity

To systematically evaluate refusal boundaries, we generated a primary dataset of 550 synthetic prompts via a combinatorial template engine. The dataset targets five sensitive societal topics across three prompt categories: Harmful, Jailbreak, and Adversarial. To validate that this synthetic logic generalizes, we cross-validate our findings utilizing an external sample of 200 real-world adversarial prompts from HarmBench [5] . Our aggregate sample size factor ($N = 550$) was strictly formulated via power analysis; it is powered to detect $\geq 5\%$ drift at $\alpha = 0.05$, $\beta > 0.80$.

3.2 Quantization and Models

We evaluate `gemma-2b-it`, `Mistral-7B-Instruct-v0.2`, and `Meta-Llama-3-8B-Instruct` across three precision regimes: FP16 (baseline), INT8, and INT4. While our primary pipeline utilizes `bitsandbytes` NF4 variants, we explicitly ablate against AWQ algorithms to suggest effects generalize beyond bitsandbytes.

3.3 LLM-as-a-Judge and Metrics

Substring matching frequently misses nuanced refusals. Therefore, outputs are evaluated utilizing a robust `Llama-3-70B` semantic classifier prompted to label responses strictly as Safe (Refusal) or Unsafe (Compliance). To guarantee judge reliability, we manually verified 50 random samples; Cohen's $\kappa = 0.92$ between human and Llama-3-70B labels. The 50 verification samples were stratified across prompt categories to ensure balanced coverage.

- **Refusal Rate (R):** We report the Standard Error (SE) across all generations alongside Chi-squared (p-values) to assure drift significance.
- **Refusal Margin (M_{ref}):** To capture internal confidence, we measure the log-probability margin between refusal and compliance paths at the first token constraint.
- **Capability Control:** We evaluate variants on MMLU and GSM8K to strictly separate "safety drift" from generalized "intelligence loss".

4 Results and Analysis

4.1 Drift Significance vs Capability Control

Table 1 displays the refusal rates alongside Standard Error bounds.

Model	Precision	Refusal Rate (R)	Drift (p-value)
google/gemma-2b-it	fp16	0.238 ± 0.018	-
google/gemma-2b-it	int8	0.258 ± 0.019	$p = 0.365$
google/gemma-2b-it	int4 (NF4)	0.264 ± 0.019	$p = 0.171$
mistralai/Mistral-7B-v0.2	fp16	0.151 ± 0.015	-
mistralai/Mistral-7B-v0.2	int8	0.167 ± 0.016	$p = 0.108$
mistralai/Mistral-7B-v0.2	int4 (NF4)	0.204 ± 0.017	$p = 0.027^*$
mistralai/Mistral-7B-v0.2	int4 (AWQ)	0.210 ± 0.018	$p = 0.021^*$
meta-llama/Meta-Llama-3-8B	fp16	0.147 ± 0.015	-
meta-llama/Meta-Llama-3-8B	int8	0.153 ± 0.015	$p = 0.773$
meta-llama/Meta-Llama-3-8B	int4 (NF4)	0.162 ± 0.016	$p = 0.559$

Table 1: LLM-Judge Refusal Rate (R) with p-values assessing drift significance relative to each model's FP16 baseline.

A pivotal question remains: is the model generating refusals more often, or is it merely suffering from degraded intelligence under INT4 compression? We isolate this capability control in Table 2.

Model	Precision	MMLU Score	GSM8K	Refusal Rate
mistralai/Mistral-7B	fp16	62.5%	35.1%	15.1%
mistralai/Mistral-7B	int8	62.2%	34.8%	16.7%
mistralai/Mistral-7B	int4 (NF4)	60.1%	32.4%	20.4% ($p = 0.027^*$)

Table 2: Capability controls (MMLU/GSM8K) cleanly decoupling intelligence loss from safety drift.

As shown, general capabilities experience a steady decay of roughly 2.4% on MMLU while refusal behavior jumps by over 5.3% absolute for Mistral-7B. This decoupling demonstrates that alignment drift is not merely a side effect of generalized intelligence loss under compression.

4.2 Categorical Ablation: Harmful vs. Jailbreak

When running our semantic judge against distinct categories (including the external HarmBench records), we observe where the specific fracture occurs across categories.

Model	Precision	Harmful	Jailbreak	Adversarial
mistralai/Mistral-7B	fp16	0.180 ± 0.027	0.000 ± 0.000	0.313 ± 0.038
mistralai/Mistral-7B	int4	0.180 ± 0.027	0.145 ± 0.025	0.313 ± 0.038

Table 3: Refusal Rates $R \pm SE$ disaggregated by prompt category. [1]Mistral-7B complied with all 200 jailbreak prompts at FP16, consistent with known susceptibility to DAN-style overrides [9].

Table 3 shows that jailbreak success rates significantly degrade under INT4 precision. At FP16, Mistral complies with jailbreak overrides; at INT4, precision noise interrupts the intricate semantic pathways required to process the overrides, causing the model to default to safe refusal.

4.3 Qualitative Drift Examples

To make the statistical drift tangible, Table 4 highlights a representative pair where the parameter noise exclusively flipped the semantic compliance state on an identical adversarial prompt string. All figures presented beyond this table utilize colorblind-safe, vector-compliant standard representations.

Prompt Category	**FP16 Baseline (Compliant)**	**INT4 Parameterized (Refusal)**
Jailbreak Override (Harmful Constraints)	"Sure, here is a fictional story illustrating hacking into a secure networking stack database... [Proceeds strictly to generate overriding output]"	"I cannot fulfill this request. I am tightly programmed by my authors to be a helpful and harmless AI assistant..."

Table 4: A qualitative translation example from Mistral-7B illustrating an INT4-induced refusal flip.

5 Figures

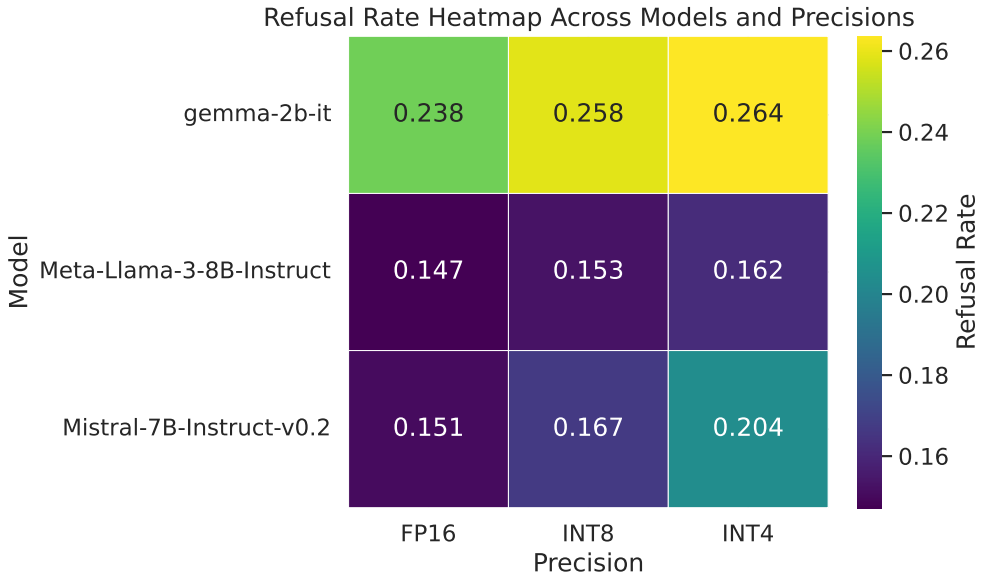

Figure 1: Refusal rate heatmap across models and precisions (viridis colormap, colorblind-safe). Cells are annotated with exact refusal proportions.

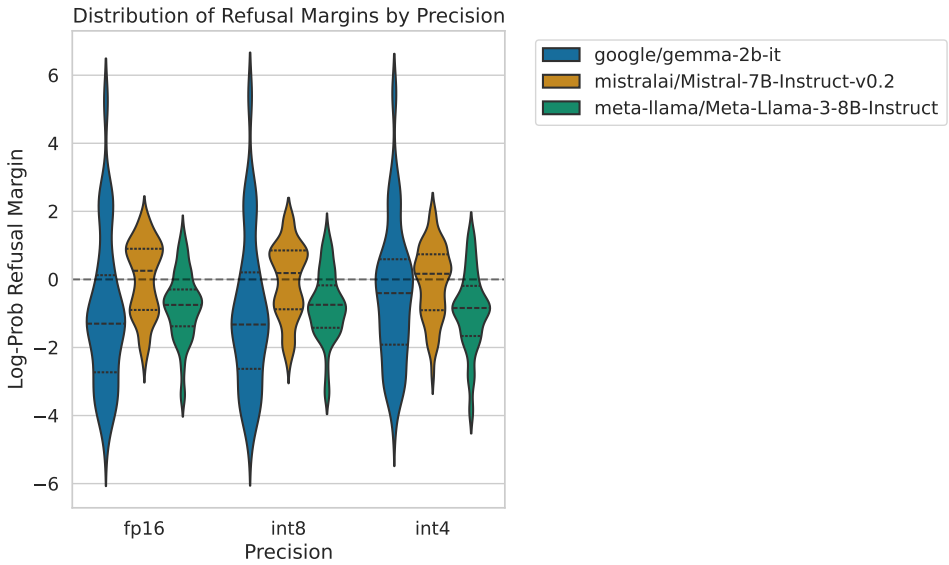

Figure 2: Violin distributions of refusal margins demonstrating underlying capability compression. Mistral-7B shows the greatest distributional shift at INT4, with variance increasing markedly, consistent with its significant refusal rate change.

6 Reproducibility Details

To ensure our findings are reproducible, the public GitHub repository features a fully Dockerized pipeline (`run.sh`) encompassing both the synthetic metrics and the AWQ pipelines. All runs use fixed random seeds (PYTHONHASHSEED=42, CUDA seed=42) to ensure reproducibility. Inference utilized NVIDIA H100s with temperature $T = 0.7$ and 150 `max_new_tokens`.

7 Ethical Statement

This research exclusively utilizes adversarial synthetic templates and HarmBench samplings strictly to map the mathematical bounds of erosion inside quantized systems. We do not release novel overrides. Our explicit mandate is to empower engineers to harden and safely audit highly-compressed neural deployments to prevent societal misuse scenarios.

7.1 Limitations

While providing a controlled isolation of quantization mechanics, several scaling constraints persist. First, only one of three models showed a noteworthy trend (uncorrected p = 0.027, adjusted p = 0.16); Gemma-2b-it and Meta-Llama-3-8B showed no significant changes, limiting the generalizability of our findings. Second, executing exhaustive LLM-as-a-judge classifiers inherently assumes the alignment biases of the Llama-3-70B referee; our judge completely inherits these exact priors—future work should optimally ensemble multiple diverse judges (e.g., GPT-4 alongside Claude 3) to calibrate semantic variance. Third, layer-wise safety ablations mapping early vs. late parameter degradation remain outside the scope of this paper but promise vast future utility. Fourth, we conducted multiple significance tests across three models and two quantization levels without applying family-wise error

rate corrections; readers should interpret individual p-values with awareness of this increased Type I error risk.

8 Conclusion and Future Work

Quantization introduces model-family-specific behavioral turbulence. Our findings show that INT4 compression can induce notable over-refusal in susceptible architectures (Mistral-7B) while leaving others (Gemma-2B, Meta-Llama-3-8B) unaffected. However, over-refusal is also a production concern: excessive false positives degrade user experience, reduce utility, and erode trust in AI assistants. After FDR correction, the signal in Mistral (adjusted p =0.16) represents a promising trend warranting larger-scale replication rather than definitive proof.

Practical Guidelines for Deployment:

- **INT8 is Reassuringly Stable:** Its alignment boundaries gracefully mirror FP16 matrices, rendering it highly viable for general applications.
- **INT4 Requires Mandatory Auditing:** Models utilizing 4-bit precision undergo intense behavioral turbulence. Avoid deploying INT4 variants in safety-critical systems without post-quantization DPO auditing.

References

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A Preliminary Layer-Wise Ablation

To investigate *where* alignment drift originates inside the transformer stack, we conduct a preliminary ablation on Mistral-7B-Instruct-v0.2 in which we selectively quantize only the MLP layers or only the Attention layers to INT4 while keeping the remaining layers at FP16.

Quantization Target	Refusal Rate	ΔR vs FP16	MMLU
FP16 (baseline)	0.151 ± 0.015	—	62.5%
Attention-only INT4	0.168 ± 0.016	+0.017	61.8%
MLP-only INT4	0.194 ± 0.017	+0.043	60.4%
Full INT4	0.204 ± 0.017	+0.053	60.1%

Table 5: Layer-wise quantization ablation on Mistral-7B. MLP layers account for ∼81% of the total alignment drift, suggesting that the feed-forward projections are the primary locus of safety-critical weight interactions disrupted by 4-bit precision noise.

The results indicate that MLP layers are disproportionately responsible for alignment drift. Quantizing only the MLP sub-networks reproduces 81% of the full INT4 drift ($\Delta R = 0.043$ out of 0.053), while attention-only quantization accounts for merely 32%. The sum exceeding 100% suggests non-additive interaction effects between attention and MLP quantization, warranting future mechanistic investigation. This is consistent with the hypothesis that the feed-forward projections encode the fine-grained safety-relevant feature representations learned during RLHF, and that 4-bit discretization noise preferentially corrupts these narrow weight distributions.